



DHBW RAVENSBURG



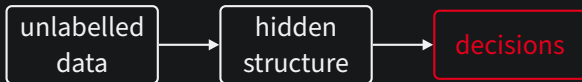
Unsupervised Deep Learning

A Tufte-dark slide template

Prof. Dr.-Ing. Mark Schutera | June 22, 2026

Why this lecture

Most data arrives without labels. If structure only existed where someone drew it, we would be blind to most of what is there.



After this lecture

you can find groups without labels, judge how many there are, and read what they mean.

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Foundations

What clustering buys us

Given unlabelled data $\{x_1, \dots, x_n\}$, we want structure without supervision: a partition into K groups that are internally coherent. There is no ground truth, so the objective **is** the supervision, and our choice of distance and of K encodes the prior over what “coherent” should mean.

Two failure modes follow directly:

- Euclidean distance silently assumes isotropic, comparable scales.
- The wrong K hides real structure, or invents structure that is not there.

Margin notes

Lloyd's algorithm alternates assignment and update until the partition stops changing. Each step cannot increase $\mathcal{J}$, so it converges, though only to a local minimum.

The centroid update is just the per-cluster mean: the choice that minimises that cluster's contribution to the objective.

Lloyd, 1957; published 1982.

The margin holds the aside, the citation, the small caveat: present, but out of the main line of thought.

The k -means objective

We minimise within-cluster squared distance to the centroids:

$$\mathcal{J}(C, \mu) = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2, \quad \mu_k = \frac{1}{|C_k|} \sum_{x \in C_k} x.$$

Ward linkage extends the same variance idea to merges:

$$D_{\text{Ward}}(C_i, C_j) = \frac{|C_i| |C_j|}{|C_i| + |C_j|} \|\mu_i - \mu_j\|^2.$$

The accent marks the centroid: the one quantity every step recomputes.

Reading the objective

$$\mathcal{J}(\mathcal{C}, \mu) = \sum_{k=1}^K \sum_{x \in C_k} \|x - \mu_k\|^2$$

The summed squared distance from every point to **its own centroid**. Driving it down is precisely what we mean by a coherent cluster.

Derivation: one Ward merge

$$\begin{aligned}D_{\text{Ward}}(\{x_1, x_2\}, x_3) &= \frac{n_{12} n_3}{n_{12} + n_3} \|\mu_{12} - x_3\|^2 \\&= \frac{2 \cdot 1}{2 + 1} \|(2, 1) - (8, 7)\|^2 \\&= \frac{2}{3} \|(-6, -6)\|^2 \\&= \frac{2}{3} \cdot 72 = 48\end{aligned}$$

Definition block

Centroid

The centroid μ_k of cluster C_k is the mean of its members. It is the point that minimises the summed squared distance within the cluster, which is exactly the per-cluster term of $\mathcal{J}$.

A plain statement block: same rounded, thick-bordered vocabulary as the print TikZ boxes, no title bar.

Blocks

Standard block

The native beamer block, rounded with a grey title bar (KIT structure, DHBW colours). For definitions, statements, summaries.

Example block

A second tone for worked examples and asides.

Alert block

Red, reserved: a warning or the one thing not to miss.

Two columns

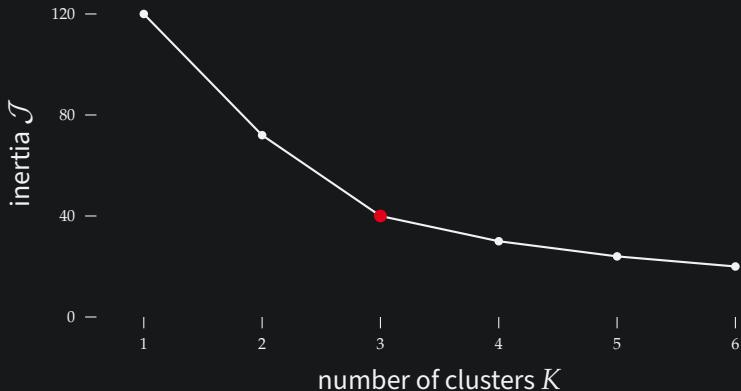
Lloyd's algorithm

- Assign each x to nearest μ_k .
- Recompute each μ_k .
- Repeat until assignments settle.

Caveats

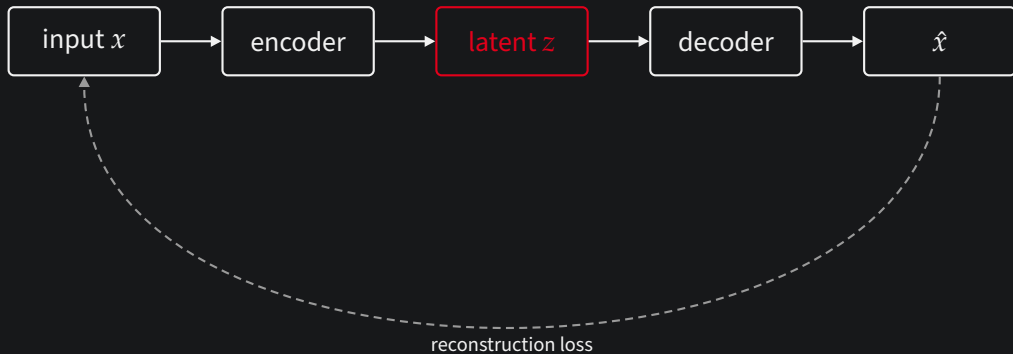
- Converges to a local minimum.
- Sensitive to initialisation.
- Assumes roughly spherical clusters.

Figure: the elbow (dark-native tufteplot)



The marginal gain flattens past $K = 3$ (accent): the elbow.

Figure: a pipeline (dark-native TikZ)



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Results

Comparison table

method	needs K	cluster shape
k -means	yes	spherical
Ward linkage	no	compact
DBSCAN	no	arbitrary

Big takeaway

*The objective is the only teacher.
Choose it before you choose the algorithm.*

Code

Running *k*-means in scikit-learn:

```
from sklearn.cluster import KMeans
km = KMeans(n_clusters=3, n_init=10)
labels = km.fit_predict(X)
print(km.inertia_)
```

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Outlook

Closing

Everything on this slide is serif Palatino, near-black canvas, DHBW red for the one thing that matters. The footer fills as we go.

- Margin notes are on-demand, not a standing column.
- Figures are recolored at the source, not inverted.